

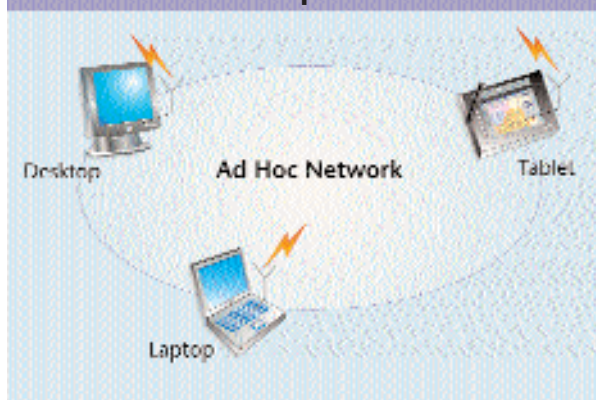
had by connecting them all together. For example, you could be talking to your buddy over a DECT (Digital Enhanced Cordless Technologies) standard cordless connection while listening to music over speakers that are connected wirelessly to an amplifier, which is controlled via your computer that you are also using to surf the Net through a wireless home gateway. All of this can be handled by one wireless network that fills the air in your house. Analyse the needs—the connection has to maintain telephone-quality sound to the DECT telephone device, while streaming real-time audio over a high priority band to the speakers in your room and handle your Web browsing at the same time. Conventional networks would be hard put to juggle this mixed traffic. Yet, HomeRF WLANs can do all this with ease.

The best part about all of this wireless business is that it's a snap to set up and use. As the end-user, you need not ever consciously twiddle with switches and configuration options—it's all automatic. Bluetooth devices find each other, exchange their capabilities and set up private connections. Laptops with wireless connectivity authenticate themselves and join the nearest open WLANs or any other computers that wish to exchange data. And with multimode adapters, your network technology hand-off occurs seamlessly from PAN to LAN to WAN and all the way back without any intervention on your part. However, as with any technology it's always good to know what's going on under the hood.

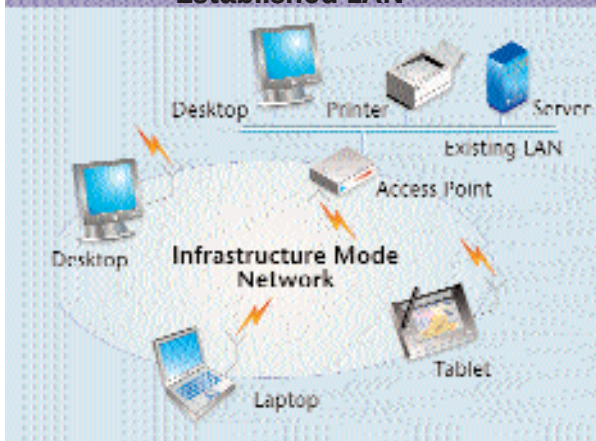
Local networking with WiFi

IEEE usually sets computer networking standards under the 802 umbrella. The 802.11 wireless standard has been around for many years, but it's only with 802.11b that it has become popular. The 802.11b standard has a range of up to 300 metres and a data transfer speed of 11 Mbps over the 2.4 GHz frequency band. Compliant devices are accorded the WiFi (Wireless Fidelity) logo by the Wireless LAN Association (WLANA). These networks work in two modes, Ad Hoc and Infrastructure. When a computer fitted with an 802.11b interface is brought within range of

Automatic Point-to-point Connections



Connecting to the Services of an Established LAN



another similarly equipped computer, both of them register each other's presence and communicate directly. A third computer will find both the existing computers and open connections to each of them separately and directly. When any of these moves out of range, it will simply be dropped from the network. This automatic, ungoverned behaviour constitutes Ad Hoc networking. If a computer is to connect with the rest of the wired world, its wireless connection must be to an Access Point. Access Points can be wired into an existing LAN, providing infrastructure connectivity to the mobile device, for services such as corporate applications, intranets and Internet access. An Access Point also behaves like a repeater, which amplifies and retransmits the network signal, doubling the range of the wireless network. Networks that include Access Points to conventional wired networks are called Infrastructure networks. Because of the way 802.11b divides its frequency range, no more than three Access Points should overlap or they will cause interference

and the performance of the network will begin to degrade. 802.11b networks also suffer from other interference problems because of the frequency band they work on.

Concerns with WiFi

WLANs work over line-of-sight, but the high-power signals can easily spill out of intended usage areas. This makes these networks rather easy to snoop on (see box, 'Drive-by Hackings') and difficult to protect. As a basic measure, WiFi includes an encryption system called Wired Equivalent Privacy (WEP). WEP encrypts each transmitted packet, but this system suffers from a few major weaknesses that render it useless against any but the most casual hackers. As a result, a new standard for security is now being deployed, called 802.1x that allows for more standard authentication schemes like Kerberos and RADIUS and greater security through proven technologies such as VPNs.

The 11 Mbps speed of 802.11b is as fast as an older co-axial cable based LAN but only one-tenth the speed of the current 100BaseT 100 Mbps networks. While that may not be blazing, it's quite adequate bandwidth for most business users. For people with higher bandwidth needs, a newer standard called 802.11a is now available, bringing speeds of up to 54 Mbps while reducing the range. Networks following this standard operate in the 5 GHz frequency band, making them incompatible with the 802.11b networks that operate over 2.4 GHz frequency band. 802.11a networks can also be packed closer; up to 12 overlapping networks can safely co-exist. While many networking companies support 802.11a, a third standard—802.11g—seeks to bring higher speed to the 2.4 GHz band, but this is not yet finalised. So there will soon be three standards vying for dominance: 802.11b (WiFi), 802.11a and 802.11g.

Quick connect with Bluetooth

Fortunately there isn't such an embarrassment of choices with PANs. As things stand today, Bluetooth rules for wireless connectivity. Bluetooth originated at Ericsson, to answer their needs for connecting handsets conveniently to a range of